

Random variables

Generally, when $f(x)$ is the pdf for

- discrete r.v. X : $\mu_X = \sum_x x f(x)$,

$$\text{Var}(X) = \sum_x (x - \mu_X)^2 f(x)$$

- continuous r.v. X : $\mu_X = \int_{-\infty}^{\infty} x f(x) dx$,

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x) dx$$

Distributions

family	mean	variance
$X \sim \text{Binom}(n, p)$	np	$np(1-p)$
$X \sim \text{Unif}(a, b)$	$\frac{1}{2}(a+b)$	$\frac{1}{12}(b-a)^2$
$X \sim \text{Norm}(\mu, \sigma)$	μ	σ^2
$X \sim \text{Exp}(\lambda)$	$1/\lambda$	$1/\lambda^2$

Sample statistics

- Sampling dist. for the mean of a sample of size n from a normal population with mean μ , standard deviation σ

$$\bar{X} \sim \text{Norm}(\mu, \sigma/\sqrt{n})$$

- For binomial $X \sim \text{Binom}(n, p)$, if $np \geq 10$ and $n(1-p) \geq 10$, we have normal approximations to sampling distributions for

$$X \sim \text{Norm}(np, \sqrt{np(1-p)})$$

$$\hat{p} = \frac{X}{n} \sim \text{Norm}(np, \sqrt{p(1-p)/n})$$

Inference Procedures

- Level C Confidence Intervals (general):

$$(\text{estimate}) \pm (\text{critical value})(\text{approx. std. error})$$

- 1-sample proportion (when $\hat{p}$ approx. normal):

$$\text{-- CIs for } p, \quad \text{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\text{-- z-test } (\mathbf{H}_0: p = p_0)$$

$$\text{test stat.: } z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- 1-sample t : test statistic when $\mathbf{H}_0: \mu = \mu_0$

$$t = \frac{\bar{x} - \mu_0}{\text{SE}}, \quad \text{SE} = \frac{s}{\sqrt{n}}, \quad df = n - 1$$

- 2-sample t : test statistic when $\mathbf{H}_0: \mu_1 - \mu_2 = 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\text{SE}}, \quad \text{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$\text{Use } t\text{-distribution with Welch } df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}.$$

- Chi-square test statistic:

$$\chi^2 = \sum \frac{[(\text{observed count}) - (\text{expected count})]^2}{\text{expected count}}$$

contingency table: $df = (\text{\#rows} - 1)(\text{\#columns} - 1)$

goodness-of-fit: $df = (\text{\#groups}) - 1$

- Model utility test:

$$t = r \sqrt{\frac{n-2}{1-r^2}} = \frac{b_1}{\text{SE}_{b_1}}, \quad \text{with } df = n - 2$$

- F -test in ANOVA: $F = \frac{\text{MSG}}{\text{MSE}}$, where

$$df_{\text{numer}} = (\text{\# of groups}) - 1, \quad \text{and}$$

$$df_{\text{denom}} = (\text{sample size}) - (\text{\# of groups})$$

Miscellaneous

- Sample standard deviation $s = \sqrt{\frac{1}{n-1} \sum_i (x_i - \bar{x})^2}$

- Conditional probability: $P(B | A) = \frac{P(A \text{ and } B)}{P(A)}$

- Bayes' Rule: $P(B | A) = \frac{P(A | B) P(B)}{P(A)}$

- Total probability: $P(A) = P(A | B) P(B) + P(A | B^c) P(B^c)$

Combinations of Random Variables

If X, Y are random variables, a, b are numbers, then

$$E(aX) = a E(X)$$

$$E(X \pm Y) = E(X) \pm E(Y)$$

$$\text{Var}(aX) = a^2 \cdot \text{Var}(X), \quad \text{or} \quad \text{SD}(aX) = |a| \cdot \text{SD}(X)$$

- Moreover, if X, Y are independent,

$$\sigma_{X \pm Y}^2 = \sigma_X^2 + \sigma_Y^2, \quad \text{or} \quad \sigma_{X \pm Y} = \sqrt{\sigma_X^2 + \sigma_Y^2}.$$

Least Squares Regression

The coefficients (from data) are given by

$$b_1 = r \frac{s_y}{s_x}, \quad b_0 = \bar{y} - b_1 \bar{x}$$